

SHETH LUJ COLLEGE

DATA VISUALIZATION PRACTICAL - 9

AIM :

a) Identify and handle missing or inconsistent data

The screenshot shows the Tableau Desktop interface with the 'Sem1_Scores' data source loaded. The 'Connections' pane on the left lists various files, including 'Sem1_Scores.csv'. The 'Fields' pane on the right shows the structure of the data source, with columns for Student ID, Full Name, Dept Info, Math, Physics, and Chemistry. The 'Data Source' pane at the bottom shows the 'Sem1_Scores.csv' file selected.

Student ID	Full Name	Dept Info	Math	Physics	Chemistry
S101	Aarav Sharma	CSE-01	85	0.900000	88
S102	priya VERMA	ECE-02	null	0.750000	80
S103	Rohan Gupta	CSE-01	78	0.820000	null
S104	Sneha Patel	IT-03	92	0.880000	90

The screenshot shows the Tableau Desktop interface with the 'Sem1_Scores' data source loaded. The 'Columns' shelf is empty, and the 'Rows' shelf is empty. The 'Marks' shelf is set to 'Automatic'. A calculation for 'Physics_Clean' is shown in the 'Marks' shelf, with the formula $[Physics] * 100$. The 'Data Source' pane at the bottom shows the 'Sem1_Scores.csv' file selected.

Physics_Clean

$[Physics] * 100$

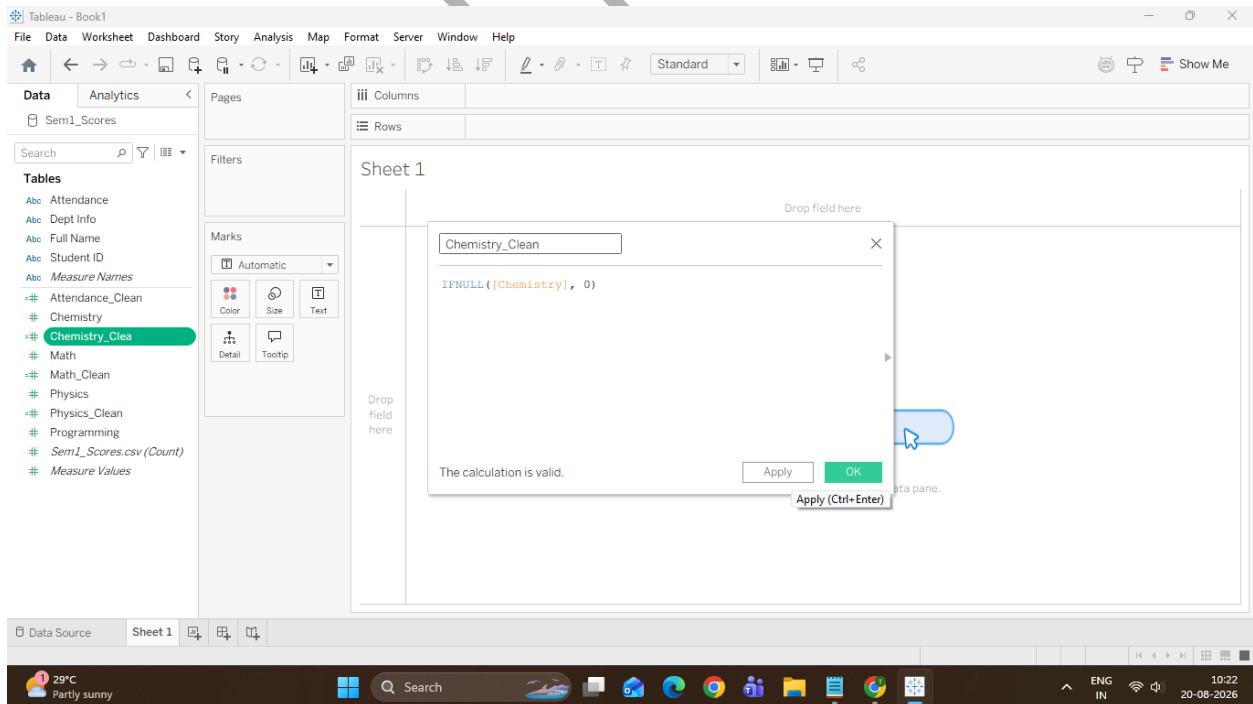
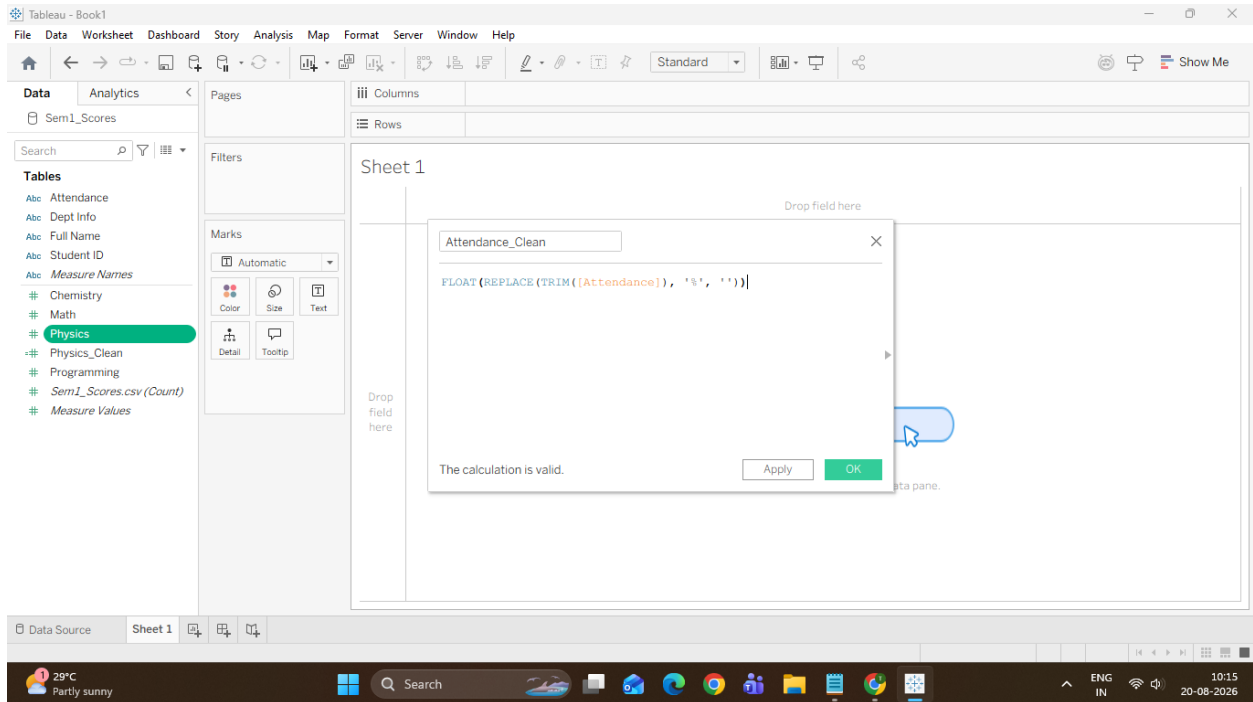
The calculation is valid.

Apply OK

DANIYAL KHAN
T088

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b) Convert wide data into tall format

The screenshot shows the Tableau Desktop interface. The 'Connections' pane on the left lists 'Sem1_Scores' as a text file. The 'Fields' pane shows the data structure for 'Sem1_Scores.csv', which has 10 fields and 40 rows. The data is organized into columns: Dept Info, Attendance, Subject, Score, and Calculation. The 'Score' column is highlighted, and the 'Calculation' column is set to 'Physics_Clear'. The 'Data Source' pane at the bottom shows 'Sem1_Scores.csv' as the selected data source.

Dept Info	Attendance	Subject	Score	Calculation
CSE-01	85 %	Chemistry	88.0000	null
CSE-01	85 %	Math	85.0000	null
CSE-01	85 %	Physics	0.9000	null
CSE-01	85 %	Programming	95.0000	null
ECE-02	90 %	Chemistry	80.0000	null
ECE-02	90 %	Math	null	null
ECE-02	90 %	Physics	0.7500	null
ECE-02	90 %	Programming	85.0000	null
CSE-01	78 %	Chemistry	null	null
CSE-01	78 %	Math	78.0000	null
CSE-01	78 %	Physics	0.8200	null
CSE-01	78 %	Programming	88.0000	null
IT-03	95 %	Chemistry	90.0000	null

c) Perform union of multiple files

The screenshot shows the Tableau Desktop interface with the 'Sem2_Scores.csv (Multiple Connections)' data source. The 'Connections' pane on the left lists 'Sem2_Scores' and 'Sem1_Scores' as text files. The 'Fields' pane shows the data structure for 'Sem2_Scores.csv', which has 10 fields and 10 rows. The data is organized into columns: Student ID, Full Name, Dept Info, Math, Physics, and Chemistry. The 'Union' dialog box is open, showing the 'Specific (manual)' tab. The 'Connection: Sem2_Scores' is selected, and the 'Tables in union: 0' is displayed. The 'Data Source' pane at the bottom shows 'Sem2_Scores.csv' as the selected data source.

Student ID	Full Name	Dept Info	Math	Physics	Chemistry
Sharma	Sharma	CSE-01	88	0.920000	90
Verma	Verma	ECE-02	79	0.780000	82
Gupta	Gupta	CSE-01	82	0.850000	80
Patel	Patel	IT-03	95	0.900000	94
Rao	Rao	MECH-04	52	0.580000	55
Singh	Singh	CSE-01	90	0.940000	88
Nair	Nair	ECE-02	74	0.730000	78
Reddy	Reddy	IT-03	96	0.960000	95
Joshi	Joshi	CSE-01	70	0.650000	62
Iyer	Iyer	ECE-02	85	0.880000	84

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Tableau - Book1

File Data Server Window Help

Connections

Sem1_Scores
Text file

Files

☐ Use Data Interpreter
Data Interpreter might be able to clean your Text file workbook.

covid_19.csv
customer_purchase.csv
Department_Master.csv
Iris (1).csv
Iris (2).csv
Iris.csv
restaurant-1-orders.csv
Sem1_Scores.csv
Sem2_Scores.csv
Student_Activities.csv
students.csv
Superstore.csv

New Union
New Table Extension

Sem1_Scores (3)

Connection
☒ Live ☐ Extract

Filters
0 | Add

Union

9 fields 20 rows

20 rows

Name	Union	Student_ID	Full_Name	Dept_Info	Math	Physics	Chemistry
Union	Union	Union	Union	Union	Union	Union	Union
S107	Rahul Nair	ECE-02	74	0.730000	78		
S108	Kavya Reddy	IT-03	96	0.960000	95		
S109	Vikram Joshi	CSE-01	70	0.650000	62		
S110	Pooja Iyer	ECE-02	85	0.880000	84		

Data Source Sheet 1

30°C Partly sunny

Search

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d) Execute cross-database joins

Tableau - Book1

File Data Server Window Help

Connections

Sem1_Scores
Text file

Files

☐ Use Data Interpreter
Data Interpreter might be able to clean your Text file workbook.

Car_Sales_T...Dataset.csv
covid_19.csv
customer_purchase.csv
Department_Master.csv
Iris (1).csv
Iris (2).csv
Iris.csv
restaurant-1-orders.csv
Sem1_Scores.csv
Sem2_Scores.csv
Student_Activities.csv
students.csv

New Union
New Table Extension

Sem1_Scores+

Connection
☒ Live ☐ Extract

Filters
0 | Add

Union

Department_Master.csv

Join

Inner Left Right Full Outer

Data Source Department_Master.csv

Search

Attendance
Chemistry
Dept_Info
Full_Name
Math
Path
Physics
Programming
Student_ID
Create Join Calculation...

Union

Name Union

Description

No description available

Fields

Type	Field Name	Physical Table	Remote Field...
Abc	Student_ID	Union	Student_ID
Abc	Full_Name	Union	Full_Name

Data preview unavailable

Data Source Sheet 1

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Search

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Tableau - Book1

File Data Server Window Help

Connections

Sem1_Scores
Text file

Files

Use Data Interpreter
Data Interpreter might be able to clean your Text file workbook.

- Car_Sales_T...Dataset.csv
- covid_19.csv
- customer_purchase.csv
- Department_Master.csv
- Iris (1).csv
- Iris (2).csv
- Iris.csv
- restaurant-1-orders.csv
- Sem1_Scores.csv
- Sem2_Scores.csv
- Student_Activities.csv
- students.csv

New Union

New Table Extension

Sem1_Scores+

1 Alert

Connection
Live Extract

Filters
0 Add

Union is made of multiple tables. ⓘ

Union

Join

Inner Left Right Full Outer

Data Source

Dept_Info = Department_Master.csv

Search

Department Name

Dept Code

Dept Target Ave

HOD Name

Total Dept Capacity

Create Join Calculation...

Name Union

Description

No description available.

Fields

Type	Field Name	Physical Table	Remote Field...
Abc	Student_ID	Union	Student_ID
Abc	Full_Name	Union	Full_Name

Data preview unavailable

Data Source Sheet 1

30°C Partly sunny

Search

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Tableau - Book1

File Data Server Window Help

Connections

Sem1_Scores
Text file

Files

Use Data Interpreter
Data Interpreter might be able to clean your Text file workbook.

- Car_Sales_T...Dataset.csv
- covid_19.csv
- customer_purchase.csv
- Department_Master.csv
- Iris (1).csv
- Iris (2).csv
- Iris.csv
- restaurant-1-orders.csv
- Sem1_Scores.csv
- Sem2_Scores.csv
- Student_Activities.csv
- students.csv

New Union

New Table Extension

Sem1_Scores+

Connection
Live Extract

Filters
0 Add

Union is made of multiple tables. ⓘ

Union

Join

Inner Left Right Full Outer

Data Source

Dept_Info = Department_Master.csv

Add new join clause

Union

14 fields 20 rows

Name Union

Description

No description available.

Fields

Type	Field Name	Physical Table	Remote Field...
Abc	Student_ID	Union	Student_ID
Abc	Full_Name	Union	Full_Name

Union	Union	Union	Union	Union	Union
Student_ID	Full_Name	Dept_Info	Math	Physics	Chemistry
S101	Aarav Sharma	CSE-01	85	0.900000	88
S102	priya VERMA	ECE-02	null	0.750000	80
S103	Rohan Gupta	CSE-01	78	0.820000	null
S104	Sneha Patel	IT-03	92	0.880000	90

Data Source Sheet 1

30°C Partly sunny

Search

ENG IN

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DATA VISUALIZATION PRACTICAL - 9

Connections

- Sem1_Scores (Text file)

Files

- Car_Sales_T...Dataset.csv
- covid_19.csv
- customer_purchase.csv
- Department_Master.csv
- Iris (1).csv
- Iris (2).csv
- Iris.csv
- restaurant-1-orders.csv
- Sem1_Scores.csv
- Sem2_Scores.csv
- Student_Activities.csv
- students.csv

Union is made of multiple tables. ⓘ

Union (dropdown) — Department_Master.csv (dropdown)

14 fields 20 rows

Name	Union	Student_ID	Full_Name	Dept_Info	Math	Physics	Chemistry
S101	Aarav Sharma	CSE-01	85	0.900000	88		
S102	priya VERMA	ECE-02	null	0.750000	80		
S103	Rohan Gupta	CSE-01	78	0.820000	null		
S104	Sneha Patel	IT-03	92	0.880000	90		

e) Restructure data for better visualization

Columns: AVG(Math), AVG(Programming)

Rows: Department Name

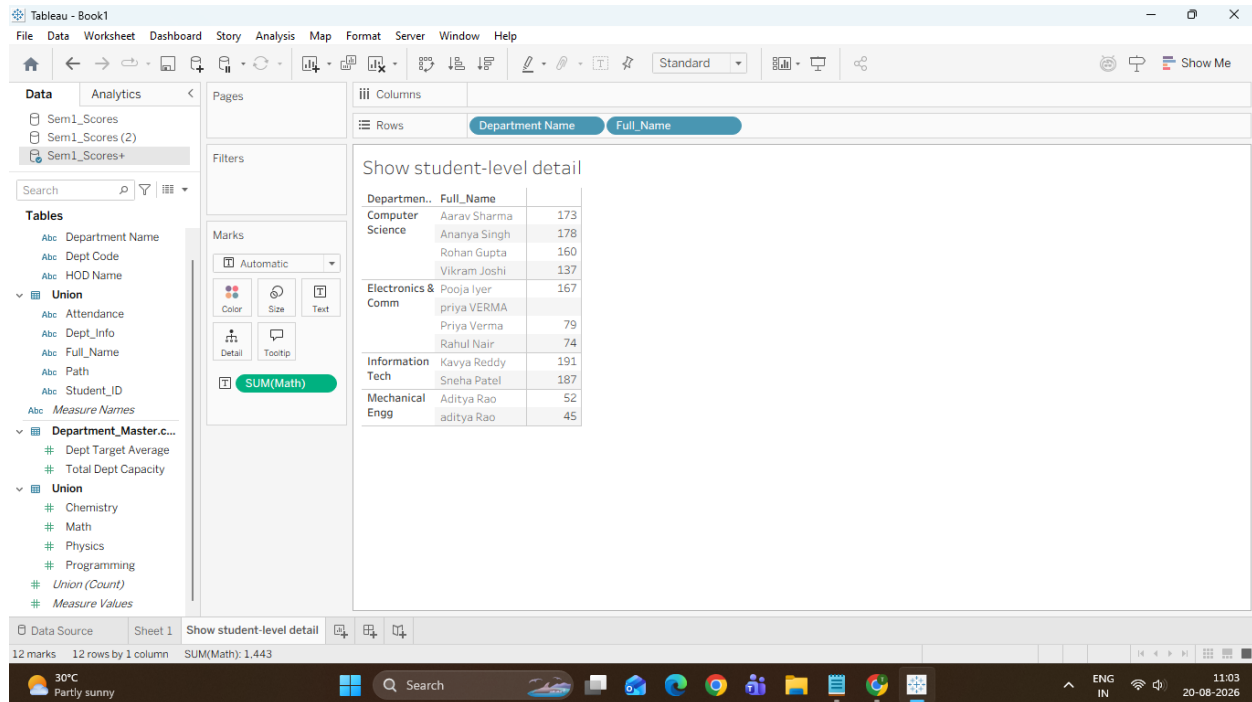
Sheet 1

Department Name	Avg. Math	Avg. Programming
Computer Science	~85	~90
Electronics & Comm	~80	~85
Information Tech	~90	~95
Mechanical Engg	~50	~65

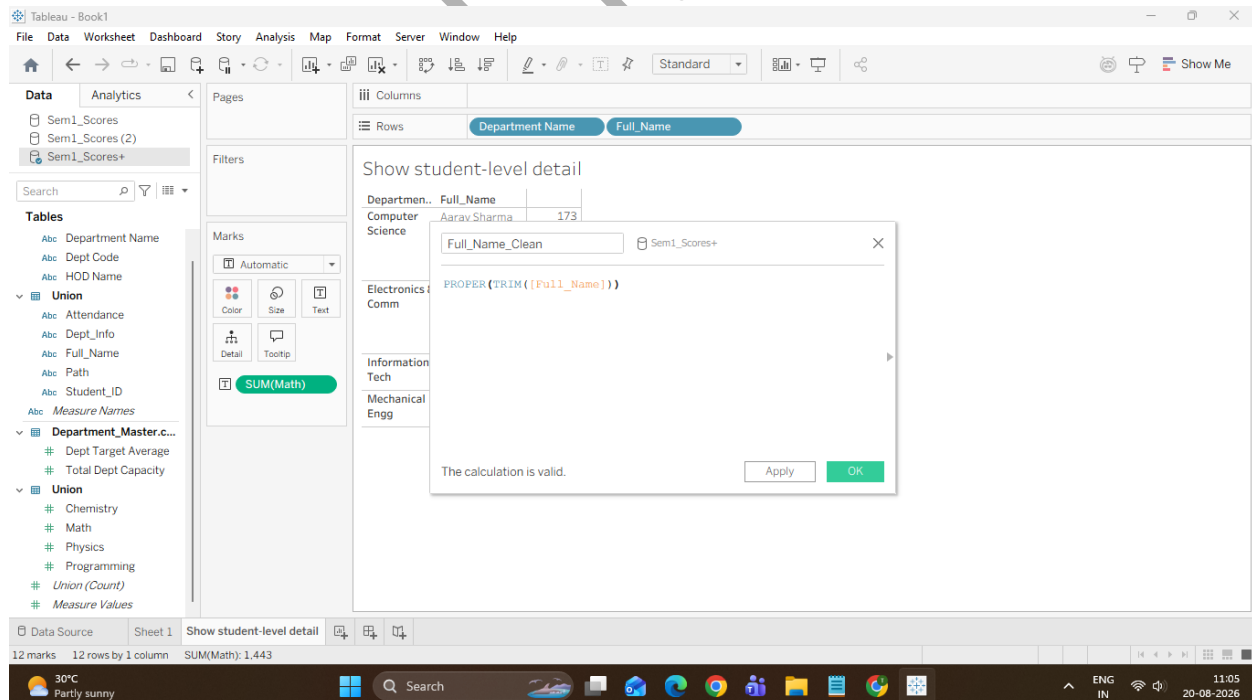
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f) Handle different levels of detail in datasets



g) Apply advanced data cleaning techniques



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